$x = \frac{22}{9}$, the required value.

We use the same exact logic to convert any number with a non terminating repeating part into a fraction.

Example 2.5 Convert the following decimal numbers in the form of $\frac{p}{q}$ $(p, q \in Z \text{ and } q \neq 0)$.

(i) $0.\overline{3}$ (ii) $2.\overline{124}$ (iii) $0.4\overline{5}$ (iv) $0.5\overline{68}$

Solution

(i) Let $x = 0.\overline{3} = 0.3333\ldots$ (1)

(Here period of decimal is 1, multiply equation (1) by 10)

$10x = 3.3333...$ (2)

(2) – (1): $9x = 3$ or $x = \frac{1}{3}$

(ii) Let $x = 2.\overline{124} = 2.124124124\ldots$ (1)

(Here period of decimal is 3, multiply equation (1) by 1000.)

$1000x = 2124.124124124\ldots$ (2)

(2)–(1): $999x = 2122$ $x = \frac{2122}{999}$

(iii) Let $x = 0.4\overline{5} = 0.45555\ldots$ (1)

(Here the repeating decimal digit is 5, which is the second digit after the decimal point, multiply equation (1) by 10)

$10x = 4.5555\ldots$ (2)

(Now period of decimal is 1, multiply equation (2) by 10)

$100x = 45.5555\ldots$ (3)

(3) – (2): $90x = 41$ or $x = \frac{41}{90}$

(iv) Let $x = 0.5\overline{68} = 0.5686868\ldots$ (1)

(Here the repeating decimal digit is 68, which is the second digit after the decimal point, so multiply equation (1) by 10)

$10x = 5.686868\ldots$ (2)

(Now period of decimal is 2, multiply equation (2) by 100)

$1000x = 568.686868\ldots$ (3)

(3) – (2): $990x = 563$ or $x = \frac{563}{990}$.

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